

Prospective Evaluation Protocol v1.0

RiskLoom · frozen 7 September 2026, before the sealed seven-day outcomes were opened

Scope. The sealed soak record: 2,016 observations, 2026-08-30T14:55:00Z to 2026-09-06T14:55:00Z, 69 symbols, audited 12/12. Evaluation uses only states RiskLoom prospectively committed and source evidence prospectively retained. No warning state, threshold, source input or observation is reconstructed or backfilled after the fact; Layer 2 outcomes are computed from the retained evidence under the predeclared rules below.

This protocol is frozen. No threshold, horizon, exclusion or definition may be added, removed or altered after the outcomes are opened. If a defect is found later it is disclosed as an amendment with its date, never applied silently.

WHAT IS BEING EVALUATED

Two layers, reported separately. They answer different questions and must not be merged.

Layer	Question	Independence
1. Structural state	What did RiskLoom observe in liquidation structure? Incidence and duration of NORMAL, ELEVATED and CASCADE_PRESSURE per symbol, and QUIET, BUILDING, ELEVATED, STRESSED at market level.	Descriptive. Makes no predictive claim.
2. Independent consequence	Did those precommitted states precede material price displacement?	Price never enters band or state computation. Materially independent of the warning construction.

Why not liquidation outcomes. Measured 7 September on historical data: the CASCADE_PRESSURE condition arithmetically entails the historical cascade spike condition — 391 of 391 slots, zero exceptions, at zero lead, because both sum the same non-negative series against the same threshold over nested windows. 98.2% of CASCADE_PRESSURE slots sit inside a cascade already in progress. In 54.5% of cases the 'subsequent' event is already satisfied by a bucket inside the warning's own input window.

Any liquidation-based outcome would measure autocorrelation and call it foresight. Excluded from Layer 2 for that reason, and the reason is stated publicly.

LAYER 2 DEFINITIONS

Material price event

From the anchor slot's closing price, the maximum **absolute** displacement reached at any point within the forward horizon, using retained Binance one-minute highs and lows:

$$\text{displacement} = \max((H_{\max} - C_0) / C_0, (C_0 - L_{\min}) / C_0)$$

C₀ is the anchor slot close. **H_{max}** and **L_{min}** are the highest high and lowest low within the horizon.

Choice	Made this way because
Extreme within horizon, not close-to-close	A move that goes 3% against a desk and reverts still happened. Close-to-close hides it.
Absolute, not directional	RiskLoom publishes no side. Scoring direction would credit it with something it never claimed.
Fixed thresholds, not volatility-normalised	A skeptic can verify 2% on BTC themselves. A normalised figure requires accepting a vol model first.

The grid — all sixteen cells reported, no winner selected

Displacement	30 min	60 min	120 min	240 min
≥ 1%	report	report	report	report
≥ 2%	report	report	report	report
≥ 3%	report	report	report	report
≥ 5%	report	report	report	report

No preferred cell. No best horizon. No optimisation of any parameter against the outcome data.

SAMPLING UNITS

Warning and baseline observations must be comparable units. One anchor per warning episode compared against every NORMAL five-minute slot would compare unlike things — and adjacent NORMAL slots overlap so heavily in their forward windows that a quiet afternoon would contribute 48 near-duplicate observations.

Warning anchors

Rule	Definition
One episode, one anchor	A continuous run of non-NORMAL state for a symbol counts once , from its first non-NORMAL slot. Not once per five-minute observation.
Anchor definition	Episode start is the first non-NORMAL slot , whether that state is ELEVATED or CASCADE_PRESSURE. A symbol may jump NORMAL → CASCADE_PRESSURE with no intervening ELEVATED slot; that episode counts, anchored at its first CASCADE_PRESSURE slot.
Escalation stays one episode	If an episode escalates ELEVATED → CASCADE_PRESSURE it remains a single episode. The maximum state reached is recorded separately, and results are reported both for all episodes and split by maximum state. The same episode must never enter two denominators as independent warnings.
Episode ends	At the first return to NORMAL. A subsequent non-NORMAL run is a new episode.
Measurement anchor	All displacement is measured from the close of the episode start slot , not from any later slot within the episode.

NORMAL baseline anchors — non-overlapping, fixed stride

For each symbol, walk its NORMAL slots in chronological order. Take the first as an anchor, then skip forward **240 minutes** before taking the next. Repeat to the end of the window.

Property	Note
Why 240 minutes	The longest horizon in the grid. Anchors therefore have non-overlapping forward windows at every horizon, reducing mechanical overlap and serial dependence between baseline observations.
Why fixed stride, not matched sampling	Matching requires choosing what to match on — time of day, volatility regime, symbol activity — and every such choice is arguable. A fixed stride is unarguable.
Sample size	Seven days, 69 symbols, one anchor per four hours yields roughly 2,900 baseline observations. Ample.
Identical machinery	Same slot grid, same symbols, same price-event definition, same horizons, same thresholds, same coverage exclusions. Not an external market statistic.

WHAT IS REPORTED PER CELL

Four numbers, always together. A ratio without an absolute difference is misleading: 1% to 4% and 15% to 60% are both 4×, and they are not equally consequential.

Figure	Definition
Warning incidence	Share of warning episodes meeting the displacement threshold within the horizon.
NORMAL incidence	Share of baseline anchors meeting the same threshold within the same horizon.
Absolute lift	Warning incidence minus NORMAL incidence, in percentage points.
Relative incidence	Warning incidence divided by NORMAL incidence.

Without the baseline the numbers are unreadable. If 68% of warning episodes see a 1% move in 60 minutes, that means nothing until NORMAL is known. If NORMAL is 65%, the warning is worthless. If NORMAL is 12%, it is a real result.

EXCLUSIONS — MECHANICAL ONLY, DECLARED IN ADVANCE

Exclusion	Treatment
Incomplete forward coverage	Forward price is recovered from klines retained at later slots, so anchors near the window tail lack full coverage at longer horizons. Such an anchor is excluded from that horizon only , and remains included at shorter horizons where coverage is complete. Applies identically to warning and baseline anchors.
Never a miss	An excluded observation is never counted as a miss or a false positive. The excluded count is reported for every cell, for both warning and baseline.
No other exclusions	No symbol, period or event may be excluded for any reason discovered after the outcomes are opened.

REPORTING BREAKDOWN

The full grid at three levels, plus per-symbol detail underneath. Aggregate figures can conceal that a result came entirely from BTC.

Level	Membership
Overall	All 69 symbols.
Majors	Frozen list, eight symbols: BTC, ETH, SOL, XRP, ADA, BNB, LINK, DOGE. This is the pre-existing RiskLoom production major set, not a classification created for this protocol. It may not be altered after the outcomes are opened.
Non-majors	The remaining 61 symbols in the 69-symbol universe.
Individual symbols	Visible but not used for claims — sample sizes will be small. Present so concentration cannot be hidden.

PROHIBITED

No parameter may be tuned against the outcome data. No cell may be selected as headline after seeing results. No statistical test may be introduced that was not declared here — no searching for whichever test yields a favourable p-value. No exclusion may be added after opening. The majors list may not be changed. Layer 1 and Layer 2 results may not be combined into a single accuracy figure.

If the results are weak, they are reported as weak. The value of this protocol is that the rules were fixed before the outcomes were visible. That property is destroyed the moment anything is adjusted afterwards, and it cannot be restored.

OUTSTANDING DISCLOSURE ITEM

thresholds.json is not in the retained evidence and is not covered by code_digest. Its digest was prospectively committed, so a third party can verify a threshold set they are given but cannot obtain one. The file must be archived and published alongside the evidence, labelled as post-soak disclosure of a pre-soak file. This is a publication requirement, not a soak failure.